Genome FASTAs

Reference databases
(CheckM, GTDB R226, GUNC)

Input

Process

Output

Stage inputs / contig variants
(Genome QC, SeqKit)

Assembly statistics
(SeqKit)

Completeness / contamination
(CheckM lineage_wf)

Composite quality score
(Genome QC)

Taxonomic classification
(GTDB-Tk, R226)

Predict rRNA genes
(barrnap)

Within-genome rRNA consistency
(BLAST+, Genome QC)

Choose one contig-length variant
(Qscore, N50, length)

tRNA prediction
(tRNAscan-SE)

Chimerism assessment
(GUNC)

Integrate measurements / status
(Genome QC)

All assessed genomes
Master QC table

ANI groups
(FastANI)

Marker phylogeny
(GTDB alignments, FastTree)

Genome atlas
(pandas, Matplotlib, seaborn)

Final selection
[optional criteria]

Explore / select / export
Genome QC Explorer

Selected subset

Atlas + linked tables
CSV / selection records